Run Multiple Operating Systems on a Single Machine with VirtualBox

Here's a complete overview of the open source tool VirtualBox that will help you to run multiple operating systems on a single machine. This article is targeted at all those who run multiple operating systems, or would like to try out new operating systems and configure them without disturbing the existing set-up of their machines. For noobs, the article contains a hands-on demo for setting up a virtual machine (VM).

Virtualisation in computer science terms means creating virtual and isolated instances of computer resources like storage, memory, processors and memory. Virtualisation exists at various levels such as at the hardware level, operating system level and application level.

Introducing VirtualBox

In very basic terms, VirtualBox is a cross-platform virtualisation manager application, which enables the user to experience various operating systems simultaneously on Intel and AMD based platforms.

Where to get it

VirtualBox is free and open source software available for free download. One just needs to start typing the word VirtualBox on a search engine and there will be thousands of links to download from, for free. Two of these are: *http://www.filehippo.com/download_virtualbox/* and *https://www.virtualbox.org/wiki/Downloads*

Installing VirtualBox on Windows

You can install VirtualBox the same way you install any other software. Once the installer is downloaded, just double-click *VirtualBox-version-win.exe* and start the installation. The installer will start and it will provide some details with regard to installation location, type of install, and the features it will install in the next few screens. A new user can safely keep going… *Next->Next* and *Next.*

Towards the end of the installation, the installer will ask for permissions to install various device drivers; choose 'Install' and finish the installation.

Additional drivers required during installation

When you install VirtualBox, you will require a few additional drivers for your virtual machines (VMs) to work and access all the hardware attached to the host.

In Windows, as soon as you start the installation on the second screen (Figure 1), you will see a list of features that the installer will install along with the set-up. Towards the end of the installation, these are the additional packages for which set-up will ask your permission.

Let's take a brief look at what these special drivers are for.

VboxUSB: This module will enable USB support inside virtual machines, and the user will be able to attach the host's USB devices to virtual machines.

VboxNetwork: This package will install two required network drivers — *vboxnetfit* and *vboxnetapp.* Both these modules are responsible for enabling networking support to the VM.